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Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

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1. Introduction

1.1 SymAware: High-level Description of the Project

SymAware develops a new conceptual framework for awareness in multi-agent systems (MASs). The robotic models and their respective specifications are integrated into the framework which produces safe simultaneous operation of collaborating autonomous agents and humans. SymAware targets to develop a comprehensive framework to support the sustainable autonomy of agents who actively perceive risks. These agents will improve their awareness and understanding through collaborative efforts between the robots and humans while completing dynamic complex tasks. The SymAware framework will be developed using compositional logic, symbolic computations, formal reasoning, and uncertainty quantification. The combined effort will characterise and support situational awareness of MAS. The project will explore research questions related to the sustainable development of awareness by learning in social contexts, quantification of risks based on limited knowledge, and formulation of risk-aware negotiation of task distributions.
These objectives will be addressed in SymAware through

1. The logical characterization of awareness using symbolic methods;
2. quantifying the symbolic reasoning for awareness with spatial and temporal ingredients for decision-making,
3. risk awareness via quantified knowledge,
4. quantifying and communicating knowledge awareness,
5. demonstrating awareness engineering in aviation and automotive use cases, and
6. identifying requirements for ethical and trustworthy awareness in human-agent interaction.

The objectives of SymAware are aligned with European Innovation Council's (EIC's) "Awareness Inside" challenge. Furthermore, SymAware extends and formalizes human-based models of situational awareness and provides a novel conceptual situational awareness framework for MASs which covers logical characterisation and integrative formal reasoning of interdependent awareness dimensions including knowledge, spatiotemporal risk and social dimensions. The developed method will support the safe transition in operations where autonomous agents and humans are involved.

In this document, we describe the high-level ideas that will be used to develop the logical framework needed in this project. This document intentionally does not go into mathematical

details of the techniques, but provides our understanding of the next steps that will be taken in the rest of the project. The concepts for logical characterization of awareness will be reviewed in Section 2. The spatiotemporal awareness for task planning will be discussed in Section 3. Aspects of risk awareness will be described in Section 4. Ethical dimension of awareness is presented in Section 5. Finally, Sections 6 and 7 will discuss awareness in the context of our demonstrators from avionics and autonomous driving.

2. Logical Characterization of Awareness

2.1 The Need for Modal Logic

It is well known that it is possible to manage and reason about knowledge by means of the logical tools of first-order logic or propositional logic, and a number of such logical frameworks have been proposed in fact, which are available in the literature. However, as pointed out by R. Fargin [1, 2], they allow us to reason with knowledge but not about knowledge. In the considered application domain we want to be able to express and check properties related to agents' beliefs about their knowledge and other agents' knowledge, which is quite different from managing queries and representing knowledge with a logical approach: it means using logic as a framework for epistemic and auto-epistemic reasoning, that is to represent agents mental states and cognitive reasoning about what they know or suppose/assume to know on themselves and the other agents.

2.2 Modal Operators for Epistemic Logic

There is not one commonly adopted definition but a number of approaches each defining the modal operators for epistemic reasoning with different syntax and/or semantics, based on the application at hand or just in order to avoid tailoring their logic to a specific one. However, based on all existing literature on this research topic it can be agreed on the following *intuitive definitions* of the main cognitive modal operators, which may allow us to better understand their meanings and avoid mistakes in defining/using them in our logic. These four main modalities are Knowledge (K, Explicit Belief (B), Implicit Belief (L) and Awareness (A). Knowledge is intuitively understandable as the modality of knowing something which is really and immutably true since we have a proof of it. Explicit Belief is instead the modality of something that we get to know as true (as given, not derived) and believe that like an act of faith, based just on the lack of evidence about its falsification. Implicit Beliefs L are the logical consequences of all our beliefs, which means that the truth of a given formula has been derived from explicit beliefs. Awareness A is the modality of conceptually (syntactically) knowing something (the dicto), the essential requirement enabling us to (later) compute and know (the re) truth or falsity of such thing.

2.3 A Driving Car Example of Awareness and Belief

Just to clarify what the semantics of awareness and belief could be in the application domain of an autonomous driving vehicle application let us consider the following example case of two cars, A and B. Suppose car A has car B in its blind spot, dangerously approaching. If we look at the world from above, it is a true fact that car B is near car A and approaching, and this could be also well known by Car B. However car A is not even aware of car B's existence and that it is really near. Instead, car A believes to have spare room to freely move around and, as a consequence, happily begins a lane change manoeuvre in the direction of car B, which ultimately causes a collision. How the reasoning framework could have helped to prevent the collision? One hypothesis could be that the logic framework should allow car B to reason and conclude with a belief that car A has not seen its presence (responsible driving) making adjustments to its position in order to be seen or get a safer distance from it. One other view could be that car A should apply a wise and long-time driver rule to be well aware that some spots of its field of view are blind, and it should not move in that direction unsafely. Of course, these properties shall be encoded as valid formulas in the KB of the reasoning framework (i.e. as rules). Note that if the reasoning framework is well designed and will correctly do its job, its presence will actually be invisible. That is, it will result in avoiding the happening of undesired events, which otherwise would have been very visible, but thankfully will never happen instead.

2.4 Logic of Knowledge and Logic of Beliefs

The basic axiomatic frameworks for logical reasoning on knowledge and beliefs are well known in the literature as the KT45 and KD45 (or S5) respectively [3, 4]. They actually differ on just one axiom, that in the Knowledge framework corresponds to stating that we must know only true things, and is actually encoded as $K_p \geq p$: if p is known then it has to be true. In the corresponding axiomatisation of belief logic, this axiom is replaced by the much weaker assumption that we don't believe in falsity, which is then encoded as the axiom $\neg L$ (false). This simple change in the set of logical framework axioms results in a really different set of valid properties between the two. It is interesting to note that due to this change of one axiom, KT45 requires Kripke models [5] whose binary relation is symmetrical, transitive and Euclidean, that is, an equivalence relation, while in KD45 it has to be serial, instead of symmetrical. That is, in any state one other state will always be reachable. This is just a consequence of the weak axiom $\neg L$ (false) put into the reasoning framework. In fact, the geometric properties of the Kripke model access relation are strongly connected with (actually mirror) the basic properties of the corresponding logical reasoning framework, and this is useful at logic design time to understand what logic axioms are required in order to make our logic exhibit exactly the properties required in our application.

2.5 Avoiding Logical Omniscience

Both the KT45 and S5 reasoning frameworks are still models of perfect reasoners: they both include axioms defining closure under implication for their modal operators, which ultimately translates into the assumption of complete knowledge and infinite resources. In fact, that axiom (plus the assumption that all propositional tautologies are already in the KB (knowledge base),

that is, our reality description model is absolutely complete) means that every truth that is not already known can definitely be derived with what is already known. Ultimately, *all* the truth is already at hand. This is an unrealistic assumption if we thought of our application as the modelling of human reasoning or of agents whose implementations clearly will not have unbounded computational resources available. Thus, it is mandatory that our logical framework shall model *non-omniscient* agents: agents with partial, *incomplete knowledge*, which may lead to deductions that may later reveal as wrong or incoherent with the knowledge base (and will be removed from it, through a so-called *non-monotonic* reasoning procedure) and with *bounded computational* resources, which is the other side of the same medal, since it implies that agents are just not able to compute the truth of things that in principle they could be able to derive from their KB. There are a number of different ways in literature to implement non-logical omniscience, which is not a peculiar problem of modal logic but a problem that shows up already in ordinary first-order logic. However, we are interested in an approach that provides a solution to avoid omniscience assumption and at the same time also supports a semantic model for modal operators, in order to allow us to implement epistemic operators.

2.6 Possible and Impossible Worlds Semantics

The standard semantic model adopted in modal logic is that of the "possible worlds", that is, the fundamental idea that besides the current state of affairs, i.e. what you think is the reality around you, there are other possible states of affairs, different but indistinguishable in one sense: whatever of these worlds you would assume as the reality, you would not have any doubt about it being the only one. This is not the only possible approach to defining the semantics of modal operators, as i.e. a logic framework of "impossible worlds" has been proposed in the literature by Levesque et al. [6, 7]. This is one main approach of the many that have been proposed in order to tackle the problem of omniscience in a logical framework. While its idea of substituting possible "worlds" with partial "situations", which are incomplete descriptions of reality allowing thus to represent uncertainty and even incoherence of partial knowledge, undoubtedly works as a model of a *non-omniscient* reasoning system, it comes however at a cost of a not intuitive approach to modelling and it deals with only depth-one formulas and with only one agent.

2.7 General Awareness Semantics

To overcome these limitations, a logic framework due to Fagin and Harper [8, 2], namely the general awareness approach to epistemic modalities, already brings a twofold value in our application domain, since allows us to model epistemic reasoning on beliefs while sticking with the common semantic of possible worlds, which at the same is very flexible in allowing to shape the logical framework in order to avoid omniscience in a variety of ways that can be tailored to our specific application domain. It is interesting to note that the complete axiomatisation of the GA (general awareness) logic can be obtained simply by adding the axiom $B_i p = L_i p \wedge A_i p$ to the KD45 framework. However, GA also goes further by giving a formal syntax and semantics for the logic of multi-agent systems in spite of a single agent and allowing nesting of the modal operators. The approach is based on a purely syntactic definition of awareness simply as a labelling function of the states, for each agent in the system, defining what that agent is aware of in each possible state.

Besides the classical axioms derived from basic propositional logic, it also defines the semantics of implicit and explicit belief as in the classic modal logic style but requires awareness of p in order to explicitly believe it. In this logic, explicit beliefs are not closed under implication (note that implicit belief still has to be it) which as a consequence allows us to model both false beliefs and also incomplete knowledge.

2.8 Including Time in the Picture

Even though the GA logic is a viable solution to model awareness, it is not necessarily the best ever or flawless, as it is improvable in a number of ways. One consideration that could be made is that time is not modelled, which is a necessity if properties related to timing or order of the events in the model have to be expressible in the considered application. In this case, temporal logic modal operators could be included in the logic framework. Of course, we are dealing only with dynamic systems, evolving along with elapsing time, but we can assume in our application the existence of a pair system, coupled to the reasoning framework, devoted to just sensing (and actuating) reality in order to translate it into inputs which produce changes in the mental state of the agents, that Is, transitions from their current world to another possible world. These changes are triggered outside the logic framework, and trigger in turn response actions from the agents. These are also implemented outside of the reasoning framework, and thus not necessarily modelled into it. In fact, explicitly modelling interactions into the logic would connote our system as a model of a communication system. This would not be necessarily wrong, but just another increase in complexity that is maybe not strictly required, as long as messages and protocols are not to be considered another subject of our reasoning framework. Even without modelling “actions” as part of the reasoning logic, it is still possible to model interactive agents’ behaviour, just like it is in a game-theory-styled approach.

2.9 An Experienced Driver Approach

Note that we just need to be sure that it is possible to discretise the interaction events so that they can be sorted out as a sequence, without introducing in this process a too coarse approximation of the modelled reality. Then, it is okay to assume agents’ (and environment) actions as an external counterpart of our agent system reasoning model. Within this view, it is right to say that in an application modelling autonomous driving, we may think of our agents not as the main drivers of the cars but as experienced co-pilots sitting aside, with the task of giving wise advice on what they should be better to do or don’t, based on their experience and understanding of what is happening. On the other hand, all actual implementation of the co-pilot suggestions, from “high level” actions down to the level of their actual translation into precise steering commands issued to the car, that is the car driving implementation, shall be a task of the main driver handling the steering wheel, completely ignored by the co-pilot.

3. Spatiotemporal Awareness in Multi-Agent Task Planning

3.1 Quantifiable Robustness Metrics for Spatiotemporal Tasks

Multi-agent systems have been successfully applied to a wide range of research domains, such as robotic teams, distributed control, resource management, collaborative decision support systems, and data mining, among others. A multi-agent system can be seen as a society of decision-making individuals, or agents, that interact by exchanging knowledge within a shared environment and negotiating with one another in order to achieve individual interests or global goals. Intelligent and autonomous multi-agent systems have the potential to perform automatic task and motion planning in accordance with complex high-level specifications such as recurrence, request-response, and time-constrained tasks. These specifications often comprise temporal and logical rules that necessitate cooperative solutions from multiple agents [9].

Temporal logics [9] provide a logical framework for expressing temporal properties, which allows for an expressive specification language. Signal temporal logic (STL) [10] is a predicate-based temporal logic that is interpreted over dense-time real-valued signals. It provides a mathematically precise language for specifying tasks and rules over continuous signals with explicit time semantics. In particular, STL allows to impose quantitative temporal properties including combinations of surveillance (e.g., "visit regions A, B, and C periodically every 30 sec while remaining in a triangular formation") and safety (e.g., "always stay at least 5 m away from moving obstacles between 5–25 sec"), among many others. Moreover, STL entails space robustness [11], a special form of the robust semantics [12] that indicate how robustly and hence how well an STL specification is satisfied. Therefore, the quantifiable space and time robustness metrics of STL can be leveraged to formulate high-level tasks for agents in multi-agent systems under spatiotemporal constraints.

By building upon the spatiotemporal robustness metrics of STL tasks, one can use the robustness level as a criterion for the motion planning and control of multi-agent systems. The team of agents is allowed to be heterogeneous, with additional dynamic couplings among them. Robustness can be considered with respect to both disturbances and the signal temporal logic tasks. In the literature [13], it is widely assumed that there exists a feasible trajectory of the entire multi-agent system that fulfils the global STL task. Given this feasibility assumption, existing works design controllers to maximize the robustness of solutions with respect to task fulfilment. When there are dependencies among local specifications, the agents are assumed to be equipped with perfect communication capabilities with respect to the task dependency relationships, and the least-violating solutions are found to treat potential conflicts in terms of

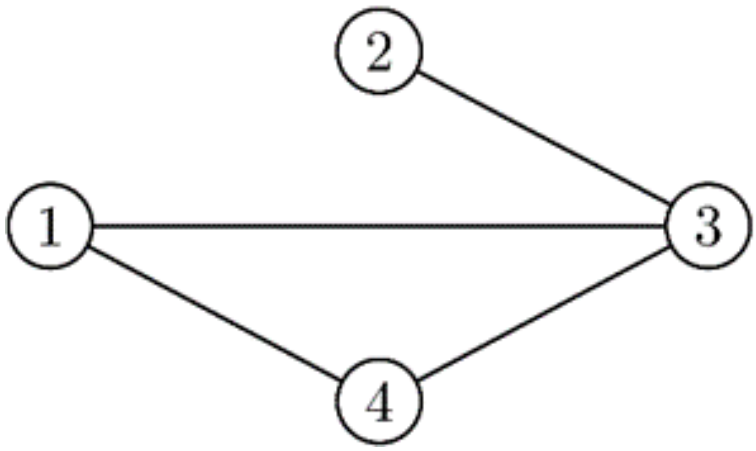

Figure 3.1: The high-level task dependency graph of a multi-agent system, with nodes corresponding to the agents and edges to the dependency relations induced by the coupling spatiotemporal logic tasks [13].

the robustness metrics [14]. However, a promising approach to dropping these assumptions is to leverage a system's awareness of the spatiotemporal task dependencies, the communication graph of the multi-agent system topology, and the robustness level of the spatiotemporal tasks [15, 16]. Specifically, one can first define spatiotemporal awareness metrics utilizing the above ingredients, and then leverage the spatiotemporal awareness of agents to design reconfiguration schemes for the motion planning of multi-agent systems, which automatically adapt to the varying environment effects caused by interaction with other agents, through evaluation of the metrics satisfactions.

3.2 Multi-agent Task Replanning under Spatiotemporal Awareness

In multi-agent systems, agents' local tasks can be interdependent, meaning that the successful execution of an agent's task may rely on the behaviour of other agents within the same subgroups or clusters. In such cases, agents should act collaboratively to ensure that all local tasks are successfully achieved. A high-level task dependency graph for a multi-agent system can be defined based on the dependency relations induced by the spatiotemporal logic specifications. For instance, Fig. 3.1 provides an example of a dependency graph for a multi-agent system consisting of 4 agents. The dependency graph is an undirected graph illustrating the relationships between agents induced by the coupling spatiotemporal logic tasks provided at the logical level. At the same time, agents use communication channels to share information with their surroundings and the other agents. However, information transmission and reception require energy, which can be a sparse commodity in many networked applications, such as sensor networks [13]. Due to the limitations in the transmission range and resolution of sensors, agents within a limited communication range can directly exchange information. The low-level communication graph represents the interaction geometry based on the topology of the information exchange between agents through active sensing and communication. Fig. 3.2 shows an example of a communication/sensing graph associated with

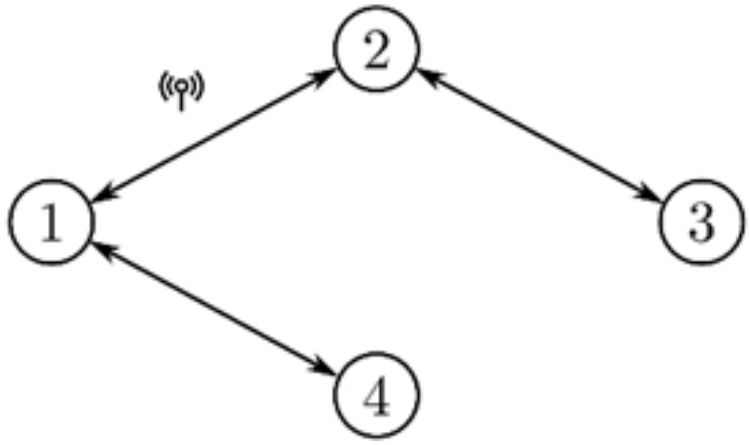

Figure 3.2: The low-level sensing graph of a multi-agent system, with nodes corresponding to the agents and edges to the interactions between agents through sensing and communication topology [13].

the multi-agent system in Fig. 3.1. The communication graph is a directed graph interpreting the allowed information flow according to the communication topology of the multi-agent system. The informational interactions between agents are dependent on the sensors constraints on vicinity visions. The edges in Fig. 3.2 suggest that the neighbouring nodes are within the sensing range of each other

As can be readily seen in the figures presented above, for a multi-agent system, the sensing and communication network may differ from its task dependency graph, which can result in infeasible spatiotemporal logic tasks. For instance, according to the high-level task graph of the multi-agent system in Fig. 3.1, the local task for Agent 1 depends not only on the states of itself but also on those of Agents 3 and 4. This indicates that coordination between Agents 1, 3, and 4 are required to accomplish the local tasks. However, the low-level sensing graph in Figure 2 reveals that Agent 3 is in fact outside of the sensing range of Agent 1, making it unable to provide any direct information to help with task satisfaction.

It is worth noting that the sensing/communication graph may be time-varying as the dynamics of the multi-agent system evolves. Therefore, it is desirable to quantify the effect caused by the mismatch between the high-level task dependency graph and the low-level sensing/communication graph and incorporate this into the robustness metrics of spatiotemporal logics. By doing so, the spatiotemporal awareness of the system can be defined accordingly with respect to the robustness level of the spatiotemporal logic tasks, and consequently used as a decision-making factor in multi-agent task planning. Specifically, when a given task specification is not achievable due to the mismatch between the high-level task graph and low-level sensing graph, the spatiotemporal awareness metric may capture an unallowable robustness value. As a result, a reconfiguration in the motion planning of the multi-agent system is needed either to find a least-violating strategy that maximizes the robustness of the original task, or to find a relaxed specification that is closest to the original one and can be fulfilled by the system.

4. Risk Awareness

For a practical system that functions in an uncertain environment, risk refers to the possibility of threats or dangerous events. A universal and generic interpretation of risk is something bad that is likely to happen but has not happened. For a multi-agent system (MAS) in which a group of robotic systems has to closely coordinate with each other and jointly accomplish a common task [17], the risk is commonly recognized as all factors that potentially prevent the achievement of the goal of the task. Coarsely speaking, risk measures the prediction of unexpected events which may affect the achievement of a certain task [18]. This also indicates that it is feasible to evaluate the likelihood of potentially unexpected events and take action in advance to prevent the occurrence in the future, which motivates, respectively, risk awareness and risk mitigation. Usually, risk awareness is the necessary condition for achieving effective risk mitigation. In practice, risk awareness for a MAS is challenging since there may simultaneously exist various risk factors that are subject to complicated dependent relations. Also, the seriousness of different risk factors may vary depending on specific scenarios, which means that different types of risk should be coped with differently. Therefore, a novel framework for risk awareness is needed to clearly and efficiently quantify risk in a MAS by precisely depicting the relation among different risk factors. A promising model for building risk awareness for a MAS is linear-time temporal logic (LTL) where risk is defined as the possible violation of one or multiple symbolic specifications written as an LTL formula [19]. By formulating the dynamic model of the MAS as a set of discrete-time difference equations perturbed by stochastic noise, risk awareness involves the quantification or estimation of the probability that this LTL specification is violated [20]. Then, risk mitigation is achieved by synthesizing a controller for the MAS with a minimal violation probability. The desired controller is expected to properly coordinate with the other agents in the environment while complying with the environmental constraints and minimizing the specification violation.

4.1 Clarifying Risks

To properly define the context of risk awareness, it is beneficial to clearly interpret risk in a specific scenario. Thus, we prefer to first introduce the notions of risk that we perceive as relevant based on the existing literature on traffic scenarios. Fig. 4.1 illustrates an essential use case of autonomous driving where each car is required to accomplish a manoeuvring task without violating the traffic rules. As an example, all vehicles in the figure are required to keep the lane while driving and overtake the vehicles ahead when needed. A speed limit sign is in the middle of the road reminding vehicles to slow down to 50 km/h before passing it.

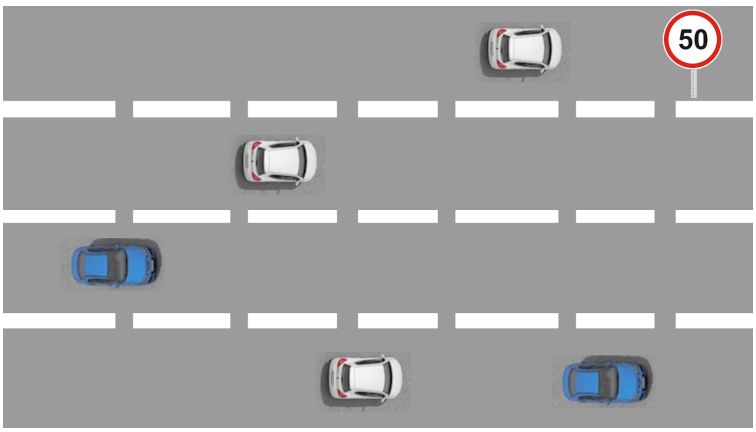

Figure 4.1: An autonomous driving use case.

4.1.1 Risk Types

In the case of autonomous driving in Fig. 4.1, risk refers to any behaviour of any vehicle that leads to the failure of the manoeuvre task or violates the traffic rules regulated by the traffic sign. In the following, we distinguish between two major types of risks, since they reflect different perspectives of risk for a MAS.

Risk of Violating Safety Requirement

The primary concern of risk for a MAS is the possible violation of safety requirements. Within a traffic scenario, a safety requirement may refer to the constraints that the motion of the agents is confined within a designated area and the distance between arbitrary two agents is beyond a minimal value [21]. Violating these safety requirements may lead to severe damage to the system and cause unrecoverable consequences. For the autonomous driving case in Fig. 4.1, common safety requirements include always driving within the lane (except overtaking) and always avoiding collisions with other vehicles. These safety constraints can be represented as a set of LTL specifications. In practice, due to the existence of environmental uncertainties, whether the safety specifications will be violated is not deterministic but is best-quantified probabilistically [22]. In many cases, there is always a chance that the constraints can be violated. In this sense, it is expected to limit the probability of constraint violation to the lowest possible level.

Risk of Failing to Accomplish Tasks

A factor that has been widely ignored is that not accomplishing a desired task expressable with LTL within a given timing horizon is also a risk. Most of the conventional methods for the design and control of a MAS mainly focus on achieving a certain goal in an infinite-timing sense but are not concerned with whether the goal is achieved within strict timing constraints. In practice, however, most jobs of practical systems are subject to strict deadlines. For the autonomous driving use case in Fig. 4.1, tasks that require finite accomplishing time include successfully overtaking a vehicle within a given time or slowing down to the desired speed regulated by the traffic sign before reaching it. Constraints with finite timing bounds are difficult to cope with due to the complexity

of the temporal descriptors. A feasible solution is to use LTL formulas with finite timing bounds to define the required specifications for a successful task [23].

The overall LTL specification of the MAS consists of the LTL formulas for both the safety requirements and finite-time accomplishments. Quantifying the violation probability of the overall LTL specification is the essential problem of risk awareness. This requires a precise prediction of the system state in the finite timing bounds and a correct quantification of the specification satisfaction. Based on this, risk mitigation is performed by solving the controller subject to the minimum violation probability of the LTL specifications. It is worth mentioning that not all risks should be evaluated in the same manner. A small risk that violates a safety requirement can be substantially more important than an average risk that does not timely satisfy a task. Additionally, within different symbolic specifications of safety one can also differentiate between the gravity of their violation. How to capture this distinguishment and incorporate it in a comprehensible way is an additional important aspect to investigate.

4.1.2 Risk Factors

Based on the essential risk awareness problem mentioned above, several factors increase the difficulty of risk awareness. Novel approaches are needed to overcome these challenges to improve the reliability and robustness of risk awareness.

Modeling and Environment Uncertainties

In practice, it is not always feasible to obtain perfect knowledge of the dynamic model of the MAS and the environment. All missing knowledge that is not incorporated by the system description is recognized as uncertainties. For the autonomous driving use case in Fig. 4.1, uncertainties include the unmodeled dynamics of the vehicles and the misperception of the sign caused by occlusions or bad weather. Various robust control methods are proposed in the literature to manage uncertainties [24, 25]. Nevertheless, most of them either rely on knowing the worst-case bounds of the uncertainties or designing self-tunable controllers, which usually leads to strong assumptions and conservative solutions. A more flexible approach is to formulate the MAS as a discrete-time stochastic dynamic model perturbed by random noise [26]. In this sense, the risk caused by violating the LTL specifications is permitted at a low probability, which achieves relaxation of the specification satisfaction incorporating the influence of the uncertainties. This provides a less conservative solution than the robust control methods since stochastic methods allow minor specification relaxation compared to worst-case uncertainties. Another effective approach is to mitigate uncertainties using learning methods [27, 28] since the system and the environment can be better known with more data during the motion of the system. For example, when the traffic sign is occluded by trees or fog, it is possible for the vehicle to obtain more details of the sign by driving closer to it. Thus, what can be realized in a practical autonomous driving scenario is to distinguish between reducible and nonreducible uncertainties and reduce the former by learning while tolerating the latter. To achieve this, certain learning structures should be designed based on proper knowledge representation which can be obtained from the work packages of Max-Planck Institute (MPI).

Risk of Human Involvement

A very important source of risk comes from humans who are involved in a human-in-the-loop (HIL) task [29]. For the autonomous driving case in Fig. 4.1, it cannot be asserted that all vehicles are driven by autonomous driving programs. Instead, an autonomous driving program should also understand human driving norms and avoid severe consequences caused by misunderstanding [30]. There are several models available in the literature [31, 32] that describe human low-level motion. High-level human behaviours are usually described using stochastic models [33, 34]. To fit the representation of environmental knowledge, human models can be organized as a set of parameterized LTL formulas that denote the human norm specifications that regulate under what condition a human will exhibit which behaviour [35]. The parameters of the LTL formulas can be learned using data collected from human-robot interaction (HRI) experiments which can be obtained from the work packages of Uppsala University. This in fact renders an inverse problem to risk awareness, i.e., inferring the LTL specifications that a human is most likely to comply with while assuming the maximum satisfaction probability.

4.2 General Formulation of Risk Awareness

The general problem of risk awareness is formulated as the analysis and synthesis of a MAS subject to a series of specifications represented as LTL formulas with finite timing bounds. The dynamic model of the MAS is in the form of discrete-time difference equations perturbed by random noise that models the stochastic uncertainties of the system. Then, the satisfaction of these specifications is quantified by probabilities. Risk mitigation should be conducted to minimize the risk probabilities within the finite timing bounds. More details are interpreted as follows.

4.2.1 Basic Risk Awareness Problem

Dynamic Model of the MAS

The dynamic model of a MAS is formulated as a Markov decision process (MDP) of which the probabilistic distributions of the state transition and observation projection are known or at least partially known. Stochastic state transitions driven by random noise represent the uncertainties of the vehicle dynamics models and the environment. Stochastic observation projection represents the possible occlusion or distortion of the vehicle sensors. To incorporate the infinite state of the vehicles, such as the positions and the velocities, discrete-time difference equations perturbed by stochastic noise are used to model the vehicles.

Risk Formulation and Quantification

The risk of safety requirement violation and task failure is formulated as the violation of a group of LTL formulas with finite timing bounds. Syntactically, these LTL formulates define the essential requirements for the MAS to accomplish the task. Therefore, risk awareness is equivalent to quantifying the probability of the MAS violating the specifications defined by these formulas. A wide class of LTL fragments can be used to model the specifications, such as LTL with finite timing bounds or signal temporal logic (STL), which will be flexibly selected to fit the properly defined tasks.

4.2.2 Risk Awareness with Additional Challenges

Risk Mitigation by Optimization: Having quantified the risk as the probability of not satisfying the LTL formulas, mitigating the risk straightforwardly leads to an optimization problem that minimizes the violation probability of the LTL formulas. To achieve this, the prediction of the system states within a certain horizon is needed to evaluate whether the system is going to violate the LTL specifications in the short future. A variety of toolsets are available to solve this problem in different manners, such as automaton-based approaches [36] and model predictive control [37]. Mixed Integer Linear Programming (MILP) [38] and dynamic programming [39] are effective tools to solve such a problem.

From Single-agent Problem to Multi-agent

One of the targets of the project is to extend the risk quantification results from a single-agent system to a MAS, which renders a nontrivial scalability issue. One critical technical point is to utilize the communication topology among the agents to address the decentralized realization of the analysis and synthesis approaches. Meanwhile, the LTL formulates also need to be adapted to this decentralized realization. From a hierarchical perspective, distributing the LTL formulas to the agents of the system as sub-specifications renders a task assignment problem or a specification distribution problem. Making such adaptations for stochastic systems is especially challenging. A new framework needs to be proposed to define and resolve specification distribution for stochastic MAS based on risk-aware negotiation among agents.

Risk Quantification based on Limited Knowledge

As an approach to reduce the system modelling and the environmental uncertainties, incremental learning can be used to allow the system to take safe action even with limited knowledge. Instead of acting only after complete knowledge is available, a more economic solution is to act immediately as the system incrementally learns the features of the environment. With a predefined structure for knowledge representation, such as a tree topology, LTL formulas are associated to regulate what the system should do under which knowledge level, rendering a flexible and responsive MAS with limited knowledge.

5. Ethical and Trustworthy Human-Robot and Human-Agent Interaction in Social Contexts with Autonomous Vehicles and Drones

An important factor in trustworthy human-agent and human-robot interaction is the person's attribution of a mind to the artificial agent or robot [40]. However, when the robot's perceived mind and actual capabilities are unaligned, this can be detrimental to a successful interaction with the person [41, 42]. Thus, it becomes very important for the robot or agent to be transparent about its abilities, relying on explainability mechanisms which would endow it with the ability to provide information about its inner workings using social cues, in a way that the human observer/interaction partner can infer how and why the agent or robot behaves the way it does [43].

In this project, we will explore new machine learning methods for the robot, in a way which would enable the human users to teach the robot explainable behaviours by providing demonstrations [44]. We will also investigate different classes of system-identified risks and safety measures in human-agent and human-robot interaction (HAI/HRI) [45, 46], develop explainability mechanisms to communicate risk and spatial information to human users, as well as design mechanisms for promoting trust and fairness and preventing bias in the developed computational methods [47]. Finally, we will develop user-centred and participatory design (PD) methods [48] and undertake user studies to investigate the ethical aspects of artificial agents collaborating with and supporting users, human-in-the-loop monitoring of the agents' interactions, and investigate adverse effects of agents' autonomy [49].

To ground the research in concrete scenarios, we will conduct the investigations above in the areas of simulation of autonomous vehicles (AV) and traffic management of multi–drone systems. The two proposed showcases will each include instances of interaction between a human and an artificial agent/robot, both as a direct interaction between the person and the AV simulator/robot/traffic management system, and indirectly during the participatory design and focus group sessions.

We plan to conduct experiments to address these main research questions:

1. How can we identify any potential ethical issues in human-agent interaction and develop mechanisms to promote fairness?

2. How can we transfer our knowledge of human goal understanding to the task of explaining

an agent's goals to humans?

3. What are the mechanisms behind indication, communication, and explanation of system-identified risk (within the context of machine operations) to another agent in an agent-human interaction?

4. How can we understand and implement the process of gathering and communicating spatial information to humans?

We translate these four questions into our research objectives, and we outline below the methodology (and the specific studies and tasks) that we will use to address them:

5.1 Identifying and Addressing Ethical Issues in Human-Agent Interaction and Mechanisms to Promote Fairness

This objective will be investigated by:

- Conducting user studies with end users in specific human-robot and human-agent interaction use cases to understand the envisioned ethical issues in the interaction;
- Investigating the concept of human oversight with user-centred design methods, researching to what extent the agent should act autonomously, how people can monitor the interaction, and how the agent could support the users without replacing them;
- Developing methods for situational awareness, following the requirement of fairness and account for inclusion and diversity, in order to avoid biases in the system;
- Investigating how to avoid gender biases and promote fairness, for example by collecting data from a balanced sample of male and female participants, at different stages of the design, development and evaluation of situational awareness technologies.

5.2 Human Goal Understanding and Explaining Agent's Goals to Humans

This objective will be addressed by:

- Developing new methods for human-agent mutual understanding, in order to enable an artificial agent to understand the intention of an interacting user and the environment;
- Building a user model to identify the relevant user goals to specific scenarios, and conducting focus groups with users to investigate what mechanisms to develop to explain agent goals to users;
- Exploring whether an "agent learning from human demonstration" approach can be used for agents to learn from users about being transparent and explainable;
- Evaluating developed methods in human-agent interaction studies to test their effectiveness on user experience and interaction quality.

5.3 Indication, Communication, and Explanation of System-identified Risk to Another Agent in the Agent–Human Interaction.

This objective's methodology will consist of:

- Using PD methods including interviews, focus groups and sketching, to develop prototypes of the explainability mechanisms to effectively communicate system-identified risk in HAI;
- Evaluating HAI studies with participants in order to measure the system's perceived trustworthiness as well as user experience and quality of interaction.

5.4 Gathering and Communicating Spatial Information to Humans

Tackling this objective will include:

- Using human-centred design methods to involve final users in the design and evaluation of methods to provide spatial information to humans.
- Conducting HAI/HRI studies with users to evaluate the effectiveness of the developed methods in terms of user experience, quality of the interaction, and user cognitive load.

6. Safety Assessment and Situation Awareness Modelling in Aviation

6.1 Safety Risk Assessment

In support of the development of aviation operations, a generic safety risk assessment cycle has been defined by Blom et al. [50] which is shown in Fig. 6.1. Step 0 in Fig. 6.1 determines the objective of the assessment and the safety context, along with the scope and the level of detail of the assessment. The complete overview of the operation is determined in step 1. In step 2, the hazards associated with the operation are recognized followed by step 3, where the hazards are aggregated into safety-relevant scenarios. Steps 4 and 5 correspond to severity and frequency assessments respectively. Step 6 helps to classify the safety risk associated with each safety-relevant scenario. The sources contributing to the lack of safety (safety bottlenecks) are recognised in step 7 while considering each safety-relevant scenario with a (possibly) unacceptable safety risk. The assessed safety risks and safety bottlenecks provide input for decision-making on operational development. The steps of this cycle are discussed in more detail next.

Step 0: Identify goal, scope, and criteria

As a basis for the safety assessment, its goal, scope and safety criteria are determined, taking into consideration user needs.

- The goal of the safety assessment is defined in coordination with the user and commensurate

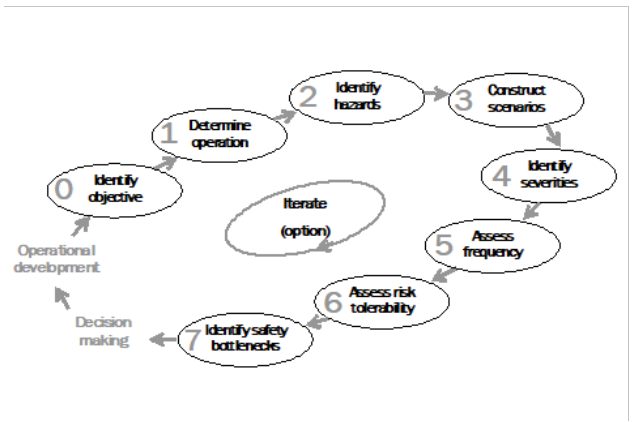

Figure 6.1: Steps in safety risk assessment cycle [50].

Table 6.1: Example of a risk matrix with criteria for acceptability of likelihood-severity combinations.

Likelihood	Severity				
	1	2	3	4	5
A	Unacceptable	Unacceptable	Unacceptable	Unacceptable	Acceptable
B	Unacceptable	Unacceptable	Unacceptable	Acceptable	Acceptable
C	Unacceptable	Unacceptable	Acceptable	Unacceptable	Acceptable
D	Unacceptable	Acceptable	Acceptable	Acceptable	Acceptable
E	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable

with the system/project life cycle and the user needs. For instance, the goal may be to provide safety feedback to operational concept designers, or to decide on allowing an operation.

- The scope of the assessment defines the boundaries of the operation, such as the boundaries of the operational area considered, and the types of functions or the types of equipment/ procedures/ people that are included. The scope also defines the types of risk that are considered in the assessment, such as all types of risk, or only particular conflict types.
- Safety criteria that are suitable for the goal and scope adopted should be identified in coordination with the user. Relative or absolute safety criteria may be used. Table 6.1 shows a typical example of a risk matrix with the absolute risk.

Step 1: Describe operation

In this step, the safety assessors obtain a complete and concise overview of the operation. The designers provide the prime input in form of a description of the operational concept. It provides an output which is sufficiently complete, structured, consistent and concise description of the operation considered and aligned with the goal and scope of the assessment.

Typical aspects that are covered in an operational concept description are: the objective of the operation; the operational context of the operation; the roles and responsibilities of the humans involved in the operation; the operational procedures; and the technical systems used in the operation.

Step 2: Identify hazards

Step 2 identifies a large number and diversity of hazards. In any condition, event, or circumstance a hazard could contribute to a safety incident or accident. In this context, both hazards that may lead to a safety relevant situation (root hazards) and hazards that may hamper the resolution of the safety relevant situation (resolution hazards) should be considered.

Hazard identification can be performed in various ways, including functional approaches, brainstorming and hazard database searches. Hazard identification is performed for a particular operational concept. If an operational concept is changed, e.g. when new safety requirements are posed, new hazards may exist and therefore additional hazard identification is needed. The hazard identification may depend on the goal of the safety assessment, and its associated level of detail. The hazard identification makes full use of the level of knowledge on the concept available

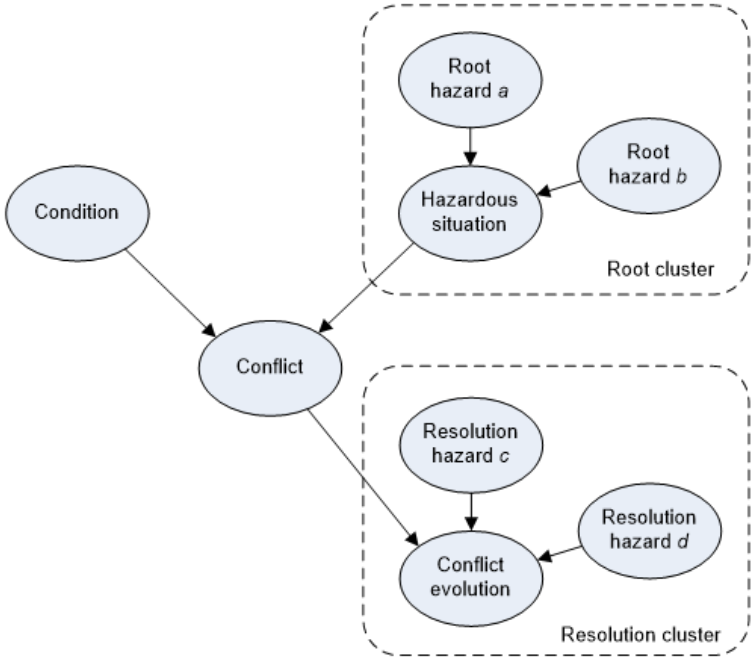

Figure 6.2: Generic diagram of conflict scenario [51].

at date. For instance, the types of hazards that are identified in an early definition phase may be at a more abstract level than those identified in a later design cycle when relations between humans/equipment/procedures in an operation are more concrete.

Step 3: Construct conflict scenarios

On the basis of the identified hazards and the operational conditions that are in the scope of the assessment, conflict scenarios (or safety-relevant scenarios) are developed. Next, the relevant scenarios that may result from the hazards are determined for each operational condition. Each potentially relevant scenario is subsequently used as crystallisation point upon which all applicable hazards and their combined effects are fitted. Next, the clusters of similarly crystallised hazards are determined to deal with the complexity of the various possible causes and results. Such clusters may play a role in several scenarios. This way of constructing scenarios aims to bring into account all relevant ways in which hazards may have a negative effect on safety in relation to the operational conditions of involved systems (such as flight phase and aircraft location) and the type of incident/accident.

Conflict scenarios can be represented in diagrams such as shown in Fig. 6.2. Examples of the elements in Fig. 6.2 for a runway incursion scenario are

- Root hazard a: The pilots react on the clearance signal for another aircraft and start crossing;
- Root hazard b: The Pilots starts to cross without clearance;
- Hazardous situation: When an aircraft is crossing the runway but it should not;
- Condition: Other aircraft has initiated take-off;

- Conflict: An Aircraft is taking off when another aircraft is crossing the runway and it should not;
- Resolution hazard c: The pilots of the crossing aircraft do not frequently look for any conflicting traffic;
- Resolution hazard d: The pilots of the crossing aircraft are not tuned to the frequency of the runway controller communication system;
- Conflict evolution: Possible ways for the evolution of the runway incursion conflict, e.g. leading to some incident or an accident.

Step 4: Identify severities

For each conflict scenario, the range of severity categories (accident/incidents) of its potential effects is identified, based on the risk classification scheme chosen in Step 0. Safety experts identify the range of potential operational effects of the conflict scenario and assess the severity levels that apply to these operational effects, using consultation of operational experts.

These operational effects may, for instance, include workload increase of operators, various degrees of separation reduction, the level of control of the situation by operators, or a collision. The effects and their respective severities depend on factors like the starting conditions and evolution of the scenario, the geometry of the scenario, and the possibilities of resolution of the conflict. Thus, a safety-relevant scenario can be instigated by a range of severities. The identification of the applicable severity classes is done prior to and separately from the assessment of the frequency of the severity categories. In relation to the risk matrix example of Table 6.1 it means identification of the columns that are applicable to the conflict scenario, while no assessment of the likelihood rows is yet done.

Step 5: Assess frequency

In Step 5, per conflict scenario for each possible severity outcome, the occurrence frequency is assessed. For example, if a particular conflict scenario could lead to either severity 1 or severity 2, depending on the circumstances, then Step 5 assesses both the frequency of severity 1 occurrence and the frequency of severity 2 occurrence. The frequency assessment can be performed in a variety of ways, including:

- Argumentation-based reasoning: Structured argumentation about operational conditions, hazards leading to conflict scenarios, the possibilities to resolve the conflict and hazards that may hamper conflict resolution;
- Fault tree / event tree bow-tie models: A combination of a fault tree, representing conditions and hazards leading to a conflict, and an event tree, representing possibilities to resolve the conflict and hazards that may hamper conflict resolution;
- Agent-based dynamic risk models: Stochastic dynamic models that represent the performance and interactions between agents in a conflict scenario, where the variability of the performance and interactions is described for nominal and non-nominal conditions.

The probability of the conflict and the conditional accident/incident probability given the conflict may be assessed by the three types of methods discussed, or by combinations.

Table 6.2: Generic risk matrix for assessment of a conflict scenario. The coloured classes are example risk assessment results.

Likelihood	Severity				
	1	2	3	4	5
A	Unacceptable	Unacceptable	Unacceptable	Unacceptable	Acceptable
B	Unacceptable	Unacceptable	Unacceptable	Acceptable	Acceptable
C	Unacceptable	Unacceptable	Acceptable	Unacceptable	Acceptable
D	Unacceptable	Acceptable	Acceptable	Acceptable	Acceptable
E	Acceptable	Acceptable	Acceptable	Acceptable	Acceptable

Step 6: Assess risk tolerability

In this step the tolerability of the risk is assessed by relating the frequencies of the severity categories of the conflict scenarios to the risk criteria. In a relative risk assessment the risk criterion is based on the implicit risk of a comparable operation (with or without some ambition factor).

The assessment of the risk tolerability should take into account the uncertainty in the assessed frequencies as determined in Step 5. As a result, for some severity levels it cannot be conclusively argued whether the risk is acceptable or unacceptable. As an example, in the generic risk matrix of Table 6.2 due to uncertainty in frequency of severity levels 1 and 3, their risk estimates cover the range from acceptable to unacceptable.

Step 7: Identify safety bottlenecks

From the risk tolerability assessment, it may follow that the risk contributions of some conflict scenarios do not satisfy the safety criteria. In step 7, the hazards, conditions and performance aspects are identified, which mostly contribute to these high risk levels. These are referred to as the safety bottlenecks. The safety bottlenecks provide operational concept designers directions for searching potential risk-mitigating measures of the operation.

6.2 Situation Awareness and Operational Safety

6.2.1 Situation Awareness Concepts

In the field of human factors, a well-known definition of situation awareness (SA) is by Endsley [52]: “Situation awareness is the perception of elements in the environment within a volume of time and space, the comprehension of their meaning, and the projection of their status in the near future”. According to the definition, SA can be determined in three levels

1. The perception of elements in the environment;
2. The comprehension of the current situation;
3. The projection of the future status.

Endsley mentioned *situation assessment* as a process of achieving, acquiring and maintaining SA [52].

In teams of interacting humans, the SA states of the humans (involved) depend on team processes. For instance, communication and coordination among the team members are needed along with cognitive processes such as perception, comprehension and projection. Team SA is defined by Endsley [52] as "the degree to which every team member possesses the SA required for his or her responsibilities". So it reflects the SA as states of team members given particular requirements for SA in the working context. Furthermore, shared SA is defined as [53] 'the degree to which team members have the same SA on shared SA requirements'. In other words, it reflects the overlap of SA elements for which team members should have SA. So, according to these definitions, team and shared SA are complex structures, which are not simply state-based, as SA is, but which can only be evaluated in the light of task-related requirements.

For joint cognitive systems, comprising humans and technical systems, the concept of distributed SA has been proposed [54, 55]. In this context SA is defined as 'activated knowledge for a specific task, at a specific time within a system', and can be held by humans and technical systems. In line with distributed cognition theory, distributed SA is achieved through coordination between the agents of the joint cognitive system, and it is viewed as an emergent property of the joint cognitive system rather than as a property of the individual agents. In a recent report on specific joint cognitive systems, involving humans and AI systems, the importance of shared SA in human-AI teams is stressed [56].

Chatzimichailidou [57] defined risk SA as the individual SA of a system agent which refers to the presence of threats and vulnerabilities that may lead to system accidents. Risk-distributed SA is considered as a special case of distributed SA, indicating that each agent, on the one hand, may have a detailed picture of the threats and vulnerabilities of the part that the agent controls, but on the other hand only retains a partial overview of the threats and vulnerabilities that are present in the entire system.

6.2.2 Mental Models

Mental models are primary means in situation assessment processes, wherein they guide attention processes and classifying information in perception (level 1 SA), they are a means for integrating the elements towards an understanding of the current situation (level 2 SA), and they are a means to reason about future states (level 3 SA). Mental models were defined by Rouse and Morris [58] as "mechanisms whereby humans generate description of system purpose and form, explanation of system functioning and observed system states, and prediction of future system states" and this definition is well in line with their role in situation assessment. Fig. 6.3 provides a schematic overview of relations between SA, mental model and overt behaviour (acting).

A review of Jones et al. [59] provides an overview of the socio-psychological literature on mental models. Mental models are personal, internal representations of external reality that people use to interact with the world around them. They are simplified representations of reality and their aspects are influenced by personal goals and motives as well as by the cultural context. In the context of risk perception, it is considered that people use mental models or scripts to reason about hazards ("anything that could lead to harm") [60]. Such mental models describe the components of a hazard, explain how a hazard may change over time, indicate who may be affected by the hazard, and explain how the hazard may be controlled.

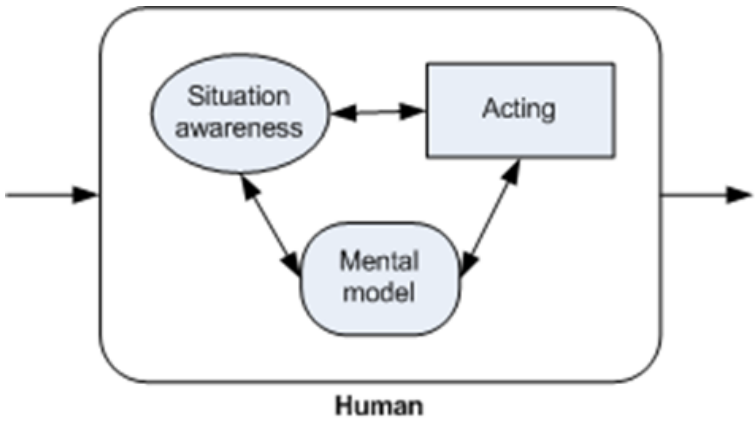

Figure 6.3: Schematic diagram of relations between situation awareness, mental model and overt behaviour (acting) of a human.

6.2.3 Risk Awareness and Risk Mental Models

As a key component of the SA of (technical/human) agents in an operational context, the agent should be aware of disturbances that impact the operation that the agent is working in, of the way that these disturbances may evolve, of the impact of these disturbances on the operations, and of the strategies that can be applied such that the effects of the disturbances are kept within acceptable bounds. Such disturbances can be any conditions, events or variations that may affect the operational performance, e.g. poor weather, high workload, failures of technical systems, or large traffic load. Part of such disturbances may have a negative effect on the safety of operations; these are named “hazards".

In choosing strategies to deal with disturbances, agents have to trade off the implications on different performance areas and on short-term and long-term goals. While safety is a key performance area, agents have to deal with the full operational complexity, and decisions impacting safety can only be understood well in relation with other objectives, e.g. completing an operation within a time frame, with acceptable costs, and with limited environmental impact. In decision-making agents have to deal with uncertainty, since the recognition of disturbances, the way that disturbances may evolve and have impact on the operations, and the effects of strategies are typically only known to a limited extent. The processes for recognizing and interpreting disturbances, and for choosing suitable strategies by individuals are based on their mental models.

Reasoning about safety is often done using the concept of risk, which combines the severity of potential consequences of a particular situation with the likelihood of attaining those consequences. Consequences considered in safety include incidents and accidents, which typically have (very) low likelihood values. The risk concept can also be used for reasoning about key performance areas other than safety, e.g. risk of financial loss, risk of continuity of operations, risk of environmental impact. Also in such areas, severity and likelihood levels can be defined, which are specific for the area considered.

Risk awareness is part of the SA of an agent and it refers to the understanding of the risks in the current work situation, notably including safety risks, and to the decisions made by the agents for strategies for dealing with the perceived risks. A risk mental model is a part of the mental model

of an agent and it describes disturbances, ways to recognize disturbances, ways that disturbances can affect particular performance areas, strategies for effectively dealing with disturbances, and ways to decide on appropriate strategies. In classic human information processing terminology, risk awareness refers to the working memory, describing risks that a person sees and understands now and in the near future, whereas a risk mental model refers to long term memory or knowledge base about risks. For example, a pilot might express his risk mental model by arguing that he would initiate a missed approach in case of an unstable approach configuration, so as to assure operational safety. The actual performance of the pilot in an unstable approach condition might be different, because he does not fully recognize the risk as the actual situation evolves and he is already having a delay.

Risk awareness can be updated by interaction processes (e.g. observation, communication, handling) with other humans, technical systems and other entities in the working context, as well as by reasoning processes using risk mental models. Risk mental models can be adapted via the SA of an individual by various learning processes, such as learning risks by own operational experiences, learning risks by communication with colleagues ("story telling"), or institutional learning about risks (e.g. training, pamphlets).

6.3 Agent-based Modelling for Safety Assessment

6.3.1 Agent-based Modelling

In order to conceptualise processes in a complex socio–technical systems it is convenient to construct an agent–oriented viewpoint. As agent-based modelling techniques study a socio–technical system as an entity comprised of several agents and the overall system behaviour emerges from individual behaviour as well as from their interactions. An example is shown in Fig. 6.4. The process furnishes a highly modular and transparent way of structuring a model for systematic analysis (both conceptually and computationally). In a complex socio–technical system the agents have boundaries which separate the internal states and processes from the external ones (environment/other agents' internal states and processes). The relationship between the internal and external states (or processes) of an agent is (strictly) represented by the inputs and outputs of the considered agent. The representation process simplifies the behaviour of complex systems having interacting entities and thereby facilitates the effective study of emergent behaviour. This makes it easier to specify models of complex systems that consist of many interacting entities, thereby facilitating effective study of the emergent behaviour of such systems.

The operational agents have the freedom to demonstrate a large variety of behavioural patterns. The characteristics of the agent and the specific processes can influence the patterns. For example, the behaviour of human agents is influenced by a wide range of cognitive and affecting aspects. These aspects can be characterised by model constructs for each agent (Fig. 6.4). However, a library of model constructs is needed in order to represent a broad spectrum of such aspects [61].

6.3.2 Multi–agent Situation Awareness Modelling

The multi–agent situation awareness model construct describes the SA of each agent (human operator, technical system) in a multi–agent system as time-dependent information of other

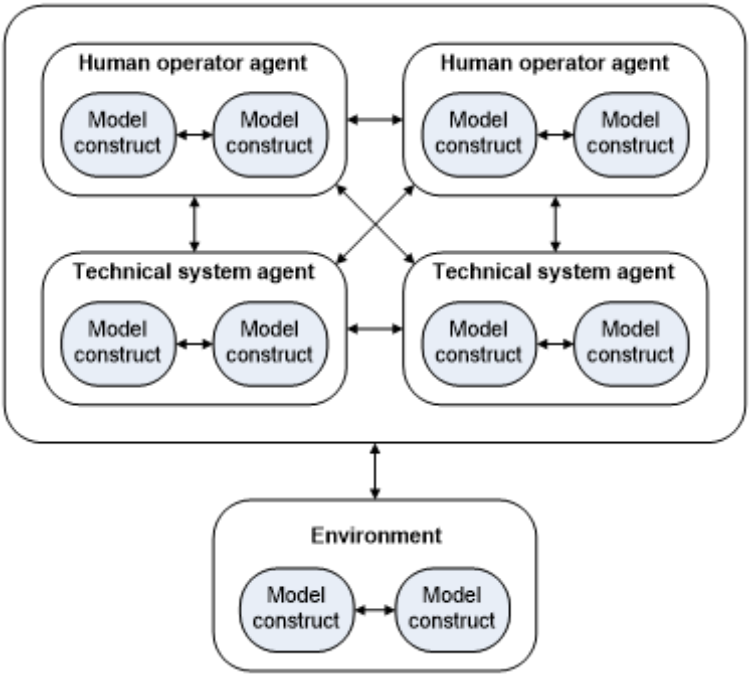

Figure 6.4: A general overview of a MAS in an environment consisting of human operators and technical systems. The relevant aspects are represented by a number of model constructs considering each agent [61].

agents, including identity, continuous state variables, mode variables and intent variables [62, 18].

In a multi–agent system consisting of n agents, each single agent may have SA about each agent and the SA of agent k at time t is denoted as

$$\sigma_{t,k} = \left(\sigma^1_{t,k}, \ldots, \sigma^j_{t,k}, \ldots, \sigma^n_{t,k}\right) \tag{6.1}$$

The SA of agent k about agent j at time t consists of various components:

$$\sigma^j_{t,k} = \begin{bmatrix} \hat{\imath}^j_{t,k} \\ \hat{x}^j_{t,k} \\ \hat{\theta}^j_{t,k} \\ \hat{\upsilon}^j_{t,k} \end{bmatrix}$$

where

- The variable $\hat{\imath}^j_{t,k}$ represents the SA by agent k at time t of the identity of agent j. For instance, the SA of a pilot regarding the identity of a nearby aircraft;
- The variable $\hat{x}^j_{t,k}$ signifies the SA by agent k at time t of continuous–valued state components of agent j. For example, the SA by an air traffic controller about the position and velocity of an aircraft;

- The variable $\hat{\theta}^j_{t,k}$ represents the SA by agent k at time t of discrete–valued state components or modes of agent j. For example, the SA of an air traffic controller about the mode of an alert;
- The variable $\hat{v}^j_{t,k}$ signifies the SA by agent k at time t of the intent of agent j. The Intent component $\hat{v}^j_{t,k}$ has multiple elements, which are discussed next.

The intent situation awareness $\hat{v}^j_{t,k}$ constitutes intent SA elements $(\hat{v}^j_{t,k})_p$

$$(\hat{v}^j_{t,k})_p = \begin{bmatrix} \bar{\theta}^j_{t,k} \\ \bar{x}^j_{t,k} \\ \bar{t}^j_{t,k} \end{bmatrix}_p$$

These elements represent the following aspects:

- $(\bar{\theta}^j_{t,k})_p$ signifies a mode of agent j that is anticipated by agent k;
- $(\bar{x}^j_{t,k})_p$ signifies a continuous–valued state component of agent j that is anticipated by agent k;
- $(\bar{t}^j_{t,k})_p$ represents the expectation of agent k, held at time t, related to the time at which the continuous–valued state component $(\bar{x}^j_{t,k})_p$ will be attained, or the expectation of the time up to which the mode $(\bar{\theta}^j_{t,k})_p$ will be attained by agent j.

A general update process for the SA is

$$\sigma_{t'+\tau} = f^{\text{SA update}}(\sigma_{t'}, \epsilon) \tag{6.2}$$

where, t' is a trigger time of the updating process, τ is the duration of the updating process and ϵ denotes stochastic effects that influence the updating process. At a high level, based on the shared information among the agents, three types of updating processes can be distinguished

- Reasoning: This is a process in which the SA of an agent is updated without considering any interactions with other agents;
- Observation: In this process, the SA of an agent is updated considering a unidirectional information flow from another agent;
- Communication: In this process, the SA of an agent is updated considering a bidirectional information flow with another agent. This can lead to a change in the SA of the other agent.

6.3.3 Rare Event Monte Carlo Simulation

Straightforward Monte Carlo (MC) simulation of the dynamic and stochastic behaviour of the interacting agents only allows assessing probabilities of safety-relevant events, such as close proximities and collisions up to a certain level in a given time. Next two MC acceleration techniques are described that can be used to assess rare event probabilities.

Interacting Particle System MC simulation

In [63, 64] acceleration of MC simulation has been achieved for air traffic scenarios using the Interacting Particle System (IPS) method of Cerou et al. [65]. It makes use of a series of decreasing miss distances. In IPS MC simulation, N runs of the agent-based model are conducted over a finite time interval in M miss distance cycles plus an additional collision cycle. In each cycle, N particles are simulated and the particles that have hit a miss distance boundary are stored. At the start of each following cycle, resampling of the stored particles is used. This means that the complete state space is copied, including all continuous and discrete. Next, the simulation of each particle is continued independently, including its unique stochastic variations as defined in the agent-based model. Finally, based on the IPS MC simulation results, rare event statistics are estimated.

Risk Decomposition for Global Failure Conditions

Risk decomposition for global failure conditions supports the assessment of rare global failures that would be encountered only very seldomly in IPS MC runs. For each global condition, a simulation setting is defined as Working, Failing, or Stochastic.

- If the simulation mode is set as Working then the related system mode in the agent-based model is set as Working during a total simulation run. This is used to evaluate the conditional risk given that the mode is in the nominal condition.
- If the simulation mode is set as Failing then the related system mode in the agent-based model is switched from working to failing at a random time during a simulation run. Next, Bernoulli sampling is used to switch back from failing to working after a mean time of the global failure mode. This scheme is used to evaluate the conditional risk given that the mode is in the failure condition during a simulation run.
- If the simulation mode is set as Stochastic then the system mode in the agent-based model is updated at each time step in the simulations according to Bernoulli sampling. As such in this simulation mode no risk decomposition is used, but rather the system mode is switched between working and failing. This simulation mode can be used if the failure mode occurs sufficiently often.

In the simulations with risk decomposition, IPS MC simulation is performed for all possible combinations of working and failing global modes. For instance, a risk decomposition for two conditions leads to conditional risks for four cases (Working-Working, Working-Failing, Failing-Working, and Failing-Failing), which are multiplied with the probability of each case, and next the risk contributions of the four cases are summed to attain the overall risk.

6.4 Use Case: Safety of Unmanned Aircraft System Traffic Management

6.4.1 Context

One of the vital elements of the current research and development (R&D) sector in air traffic management (ATM) is to assure a safe integration of unmanned aircraft (UA) in all airspace

classes, including urban areas. A key objective of the European development is U–space [66], which represents a set of new services and specific procedures which are designed to support safe, efficient and secure access to airspace for large numbers of unmanned aircraft systems (UASs). These services rely heavily on automation and digitization of tasks, whether they are carried out by the drone itself or in the context of the ground environment.

The effectiveness of new concepts and technologies must be evaluated in terms of whether they function as intended, as well as the hazards that result from malfunctioning components of those concepts and technologies. This means that two different sorts of questions must be addressed for the safety evaluation of new UAS/UAM and U-space-enabled operations:

- When systems are working as intended in normal operating conditions, how effective are detect-and-avoidance (DAA) systems, and how may they be tailored most effectively? What effects do traffic density and airspace planning have? What effect do typical sensor mistakes have? What effect do typical communication delays have? What effect do typical human reaction times have? What effect does typical variability have on the rates of operations? What effect does typical weather variation have? Etc.

- When failures or off-nominal operating conditions occur, what effects do malfunctions in technological systems, such as drone propulsion, communication, surveillance, navigation, and DAA systems, have? How effective are mitigation strategies for failure scenarios? What are the consequences of human errors, such as mistakes made during planning or in responding to DAA advisories? What effects may unexpectedly bad weather have? Etc.

Answering these types of questions in quantitative safety risk assessment of UAS/UAM operations requires modelling and simulation approaches that can represent the systems, human operators and environment of relevant operational concepts. The models should represent the performance and uncertainty of normal operations, as well as the impact of off-nominal/failure conditions that may affect the operations. The simulations should effectively represent the interactions of all entities and their variability in normal and off-nominal/failure conditions, such that probabilities of rare safety events can be estimated in a reasonable time.

6.4.2 Agent-based Model of UAS/UAM Operations

An agent-based model for safety risk assessment of UAS/UAM operations was developed by Stroeve et al. [67]. A high-level overview of the agent-based model is provided in Fig. 6.5. It shows a number of unmanned aircraft that reside in a particular environment and airspace design. The UA movements are influenced by weather conditions and the weather is being forecasted by meteorological services. There are customers who pose demand for drone missions. UA operators develop flight plans based on the mission demand, weather forecast and airspace layout. The flight plans are used by pilot in command (PICs) as a basis to control the drone missions. The mission control commands by each PIC are uplinked by the C2 link to the flight management system (FMS) of the UA and this determines the flight performance (movements of the drone). The FMS also uses ownship state estimation data, consisting of global navigation satellite system (GNSS)-based position and speed estimates and pressure altimetry. This ownship data is shared with other UAs via dependent surveillance. The ownship and othership states data are used by the DAA system on board of each UA to generate alerts and guidance advisories. The DAA output is downlinked by a command & control (C2) link and shown on the traffic display in the

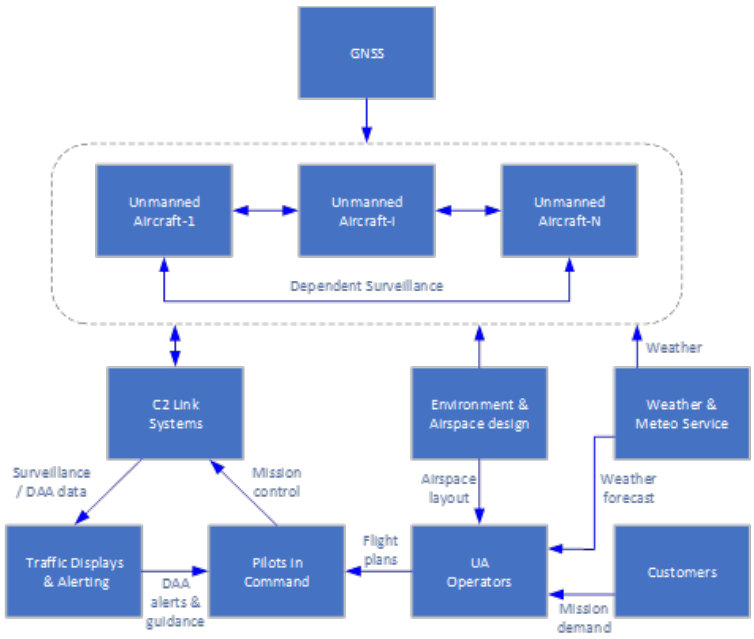

Figure 6.5: High–level overview of the agent-based model [67].

remote pilot system to the pilot in command (PIC). In response, the PIC can adapt the mission to avoid close proximity to another drone. In the agent-based model of the UAS/UAM operations, the state estimates and decision rules of all technical systems and the situation awareness of the human operators are represented. The dynamics and interactions of the agents lead to emerging behaviour in the overall modelled operations, including the occurrence of safety-relevant events. It is important to depict performance variability in both nominal and off-nominal conditions for simulation-based safety risk evaluation of operations. For this purpose, the agent-based model includes a range of discrete states that describe nominal and off-nominal modes that influence particular characteristics of system performance. The system performance at a particular aircraft i may be influenced by local systems at the aircraft as well as by global systems outside of the aircraft (which also influence systems of other aircraft).

A schematic diagram of these elements of an agent–based model is shown in Fig. 6.6. It shows two characteristics for local and global systems, which all have a nominal mode Working and an off-nominal mode Failing. Transitions between the modes are via delay gates, where the delays are chosen from exponential distributions. For instance, this model structure can describe the performance of GNSS-based state estimation at an aircraft. Here, characteristic A may represent availability for a local system (GNSS receiver at aircraft i) and for a global system (GNSS performance in the region). Both the local and the global systems need to be in the Working mode for GNSS-based state estimation at aircraft i. Lack of GNSS-based state estimation implies that position estimates are not available for navigation and surveillance functions. Characteristic B in Figure 6 may represent the level of accuracy of GNSS-based state estimation as achieved by the local and global systems. Here the Working mode may represent a normal accuracy level, where state estimates are affected by normal errors, and the Failure mode may represent a reduced accuracy mode with enlarged errors in the state estimates. The achieved accuracy affects

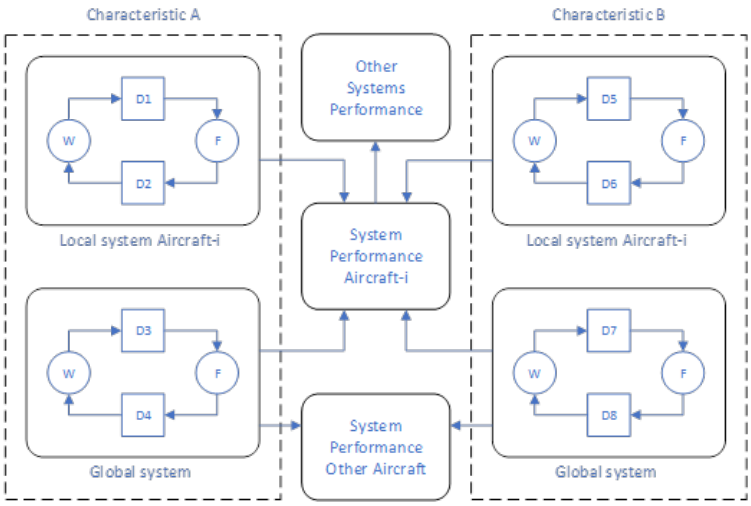

Figure 6.6: Schematic diagram of Working and Failing modes of local and global system characteristics A and B influencing system performance of aircraft i. D1 to D8 are transitions with exponentially distributed delays [67].

the navigation and surveillance performance of the affected aircraft.

6.4.3 Illustrative Simulation Results

The performance of the PIC in responding to the downlinked DAA advisories is an important factor in the effectiveness of the joint cognitive system consisting of the DAA system and the PIC [67]. The model of the PIC performance includes models for the delay in response and for the response mode. For a sensitivity analysis two settings are used for the response delay:

- Slow: mean delay is 9 s and the standard deviation is 3 s;
- Quick: mean delay is 3 s and the standard deviation is 1 s.

The response mode makes a distinction between a mode where the PIC responds to a DAA advisory and a mode where the PIC does not. The PIC can respond in one of three ways, which are examined in a sensitivity analysis:

- Altitude Response: The PIC responds to DAA altitude guidance only;
- Direction Response: The PIC responds to DAA direction guidance only;
- Both: The PIC responds both to direction and altitude guidance.

The PIC response choices for close proximity incidents are shown in Fig. 6.7 for UAS en-route crossing operations. The risk reduction factor is calculated by considering the close proximity probability for a scenario without DAA divided by the close proximity probability for a scenario considering DAA and a PIC response option.

It follows that the largest risk reduction is achieved when the PIC responds to both the direction and altitude guidance with a small delay. It is more effective to only alter altitude than to only

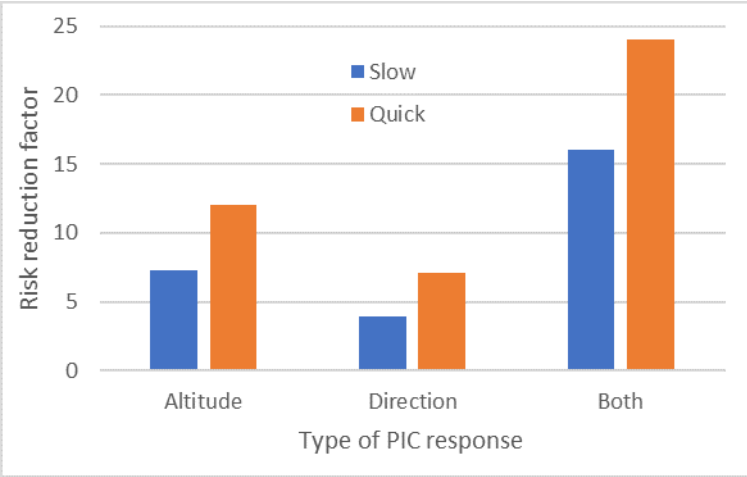

Figure 6.7: Close proximity risk reduction factors for types of PIC responses to DAA advisories in UAS en-route crossing operations: Altitude response only, direction response only, both altitude & direction response, slow and quick response [67].

change direction if the PIC would only react to one of the guidance dimensions. These types of results provide safety feedback to design and they can provide a basis for the development of operational performance standards.

7. Unknown-unsafe Scenario Creation for Safety Assessment

Even with millions of miles of recorded data, you might not encounter the edge cases a L3+ automated vehicle should be able to cover. Some scenarios are too unsafe for a test vehicle to record in the real world. How to address those gaps? For L3+ systems it becomes increasingly challenging to understand if all scenarios have been covered and the main question is how do you know what you don't know?

Traditionally, safety assessment of vehicles is mainly restricted to functional safety, as described in ISO 26262. This focuses on hazards induced by technical failures, which can be caused by systematic and random faults in hardware and software. By contrast, ISO/PAS 21448 defines Safety of the Intended Functionality (SOTIF) focussing on an intended function that could induce hazards due to functional insufficiencies, in the absence of technical system failures or by reasonably foreseeable misuse by persons. SOTIF analysis covers identification of system weaknesses as well as *scenarios that lead to a hazardous event*. SOTIF standard categorized scenarios in four areas based on two indicators (See Fig. 7.1). One for criticality and one for novelty. Main objective of SOTIF is to decrease area 3 by identifying unknown-unsafe scenarios.

7.1 Approach to find Unknown-unsafe Scenarios

Siemens proposes a four-phase solution starting from data collection, Basic Scenario elements extraction, massive scenario generation and optimization towards unknown-unsafe scenarios (see Fig. 7.2).

Data collection: for data collection we focus on different scenes with different features. For each scene, the data collection consists of two parts: roads and traffic. The road information will be measured and it will be converted to an OpenDrive file for easy integration with the simulation environment. For the traffic data, all traffic on the scene will be recorded storing the path, velocities and actor types (e.g. car, truck or pedestrian). The coverage of the traffic data will be checked to make sure the collected data represent variations in behaviour of the road users on that specific scene. The collected traffic data will be converted to OpenScenario format, thus enabling further processing (extraction) as well as importing it into a simulation environment.

Extraction: a graph-based approach is used to extract the Basic Scenario Elements from the trajectories of the road users measured in the data collection phase. Entrance and exit gates, defining the boundaries of the scene, are generated based on the OpenDrive file and used to construct the graph. Then the OpenScenario files generated in the data collection phase are all

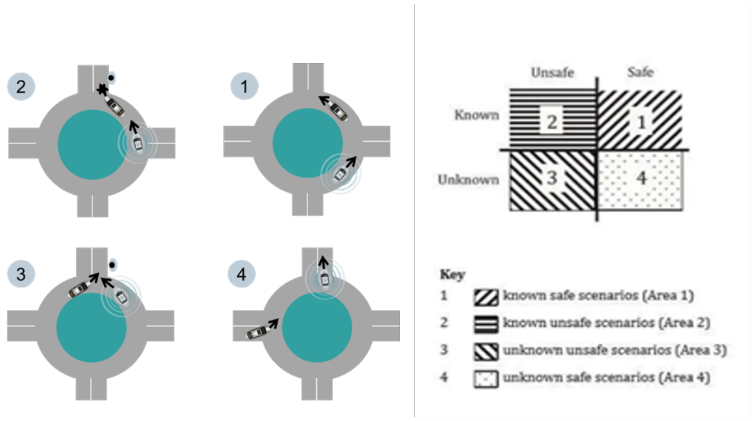

Figure 7.1: SOTIF scenarios.

linked to the entrance and exit gates defined, grouping all data into Basic Scenario Elements with each a specific entrance gate, exit gate and actor type category. Then, for each Basic Scenario Element, from all related OpenScenario data an average path and speed profile is calculated, as well as the probability distribution for several parameters defining the variations around this average path and speed profile.

Generation: in phase 3 a model is generated allowing to run a massive number of possible scenarios for all road users based on the graph and any combination of Basic Scenario Elements and the defined parameters. Based on the probability distributions defined, for each actor a probability (P_{actor}) can be calculated as the product of the individual parameter probabilities (P_{param}) The total probability of any concrete scenario executed ($P_{scenario}$) can then be calculated as the product of the probabilities of the relevant actors.

$$P_{actor} = \prod_{i=1}^{n} P_{param,i}$$

$$P_{scenario} = \prod_{j=1}^{m} P_{actor,j}$$

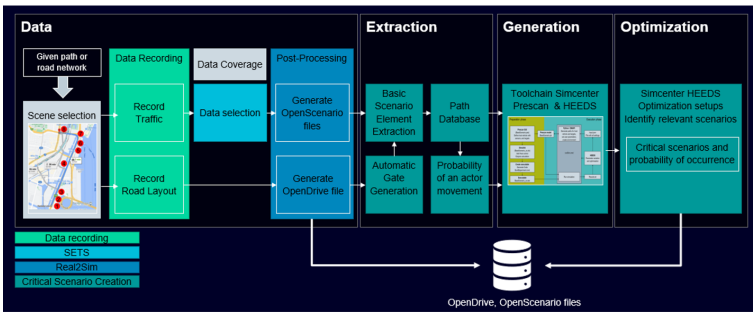

Figure 7.2: Critical scenario creation framework.

The model also defines metrics for novelty and criticality, to identify to what extent a specific

scenario is unknown/unsafe.

Optimization: in Phase 4, a multi-objective optimization algorithm is used to find unknown-unsafe scenarios, by running variations of the model defined in the generation phase. The objective in the optimization is based on novelty, criticality and scenario probability. Thanks to Simcenter HEEDS the search space can be explored with a limited number of test cases. The result will be a collection of various different unknown-unsafe scenarios.

8. Conclusions

This document described the high-level concepts and ideas that will be used to develop the logical framework needed in the SymAware project. We intentionally did not go into mathematical details of the techniques, but provided our understanding of the concepts that will be used in the rest of the project. We described the concepts for logical characterization of awareness, the spatiotemporal awareness for task planning, risk awareness, and ethical dimension of awareness. We plan to refine the content of this document throughout the project as research results are becoming more concrete.

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